

Supplementary Material

Wire-Level Interrupt-to-Decision Latency of On-Sensor MLC versus Host Inference on the NVIDIA Jetson Orin Nano: A Pre-Registered Measurement Study

Not part of the 4-page Letters build (`build_tex.py` does not read this file). S2 and S4 are referenced from the main text by short pointer sentences (§V.A and §VI.A, respectively). S1 and S3 are provided here without an in-text manuscript pointer, per the page-budget decision made during this revision.

S1. Full percentile table (extends Table I)

Table I in the main text reports only median and IQR per pipeline×condition cell, to fit the page budget. The full table, including p95, p99, max, and mean, is reproduced here.

Wire-Level Latency per Pipeline × Condition (μs)

Latency = $t(\text{D1 rising}) - t(\text{D0 rising})$. Each cell aggregates 9 blocks of 300 s. n = included trials (4,770 of 4,860 candidates, 98.15%). IQR = interquartile range (p25–p75). All values μs except n . p95/p99/max computed via linear interpolation (numpy default); see `code/analysis/compute_percentile_table.py` and `data/processed/confirmatory-2026-05-26/percentile_table.csv`.

Pipeline	Condition	n	Median	IQR	p95	p99	Max	Mean
host	idle	536	321.7	319.7–326.3	347.3	506.3	642.9	328.8
host	i2c-cont.	529	574.5	547.8–599.6	639.5	662.3	854.1	570.4
host	stress	532	345.0	342.0–348.9	361.2	400.6	3504.8	351.3
mlc	idle	525	681.5	505.4–1086.7	179.7	2120.6	2604.5	866.6
mlc	i2c-cont.	532	1325.4	1283.7–1376.3	136.2	1675.4	1904.8	1333.2
mlc	stress	527	546.1	535.9–557.6	579.5	879.1	3868.2	560.6
mlc-binary	idle	531	231.9	61.8–246.2	485.3	654.2	723.9	236.7
mlc-binary	i2c-cont.	527	49.4	46.9–53.1	246.7	272.1	289.3	64.5
mlc-binary	stress	531	70.2	66.9–73.0	78.4	197.5	253.7	72.0

Note: The mlc/idle mean (866.6) exceeding its median (681.5), and the wide mlc/idle IQR, reflect the multimodal structure discussed in §V.B. The mlc-binary pipeline performs zero I²C transactions on the decision path, isolating the kernel/gpiod latency floor.

S2. Host vs. MLC platform power (post-hoc; referenced from §V.A)

Jetson platform power (VDD_CPU_GPU_CV, INA3221) differs between host and MLC pipelines by +31.1 mW [27.8, 34.4] at idle ($p = 8.4 \times 10^{-2}$) and +7.3 mW under I²C contention ($p = 5.5 \times 10^{-2}$). This comparison was added post-hoc in response to review; per the pre-registration’s standing rule (v7.13), it is labeled as such rather than presented as part of the original confirmatory design. Under CPU stress, the 95% CI spans [−28.8, 14.6] mW and includes zero despite $p = 1.4 \times 10^{-2}$ — expected at this sample size, where MWU is sensitive to distributional differences beyond central

tendency; the CI, not the p-value, is the practically relevant statistic here. VDD_IN replicates the idle effect at +37.3 mW [33.2, 41.4]. These are whole-platform measurements, not an isolated host-classifier-vs-MLC comparison — the sensor’s own power draw is not separately instrumented.

S3. Candidate mechanisms for the idle multimodal distribution

We do not present a confirmed mechanism. Candidate explanations include: (a) idle-state I²C bus arbitration on the Jetson’s `i2c-tegra194` driver having multiple equilibrium timings between conflicting wake-up paths, (b) the `gpiod` write path through `/dev/gpiochip0` traversing different kernel call sequences depending on whether the underlying chardev poll mechanism is in steady-state or recently-armed, or (c) some interaction between the MLC’s 706.5 ms internal cadence (§V.C) and the kernel’s microsecond-resolution interrupt-arrival timing. Distinguishing these requires `ftrace` instrumentation of the I²C driver and the `gpiochip` event flow at the kernel level, which lies outside the scope of this wire-level study.

S4. Measured SDA/SCL capture of the three-transaction read (referenced from §VI.A)

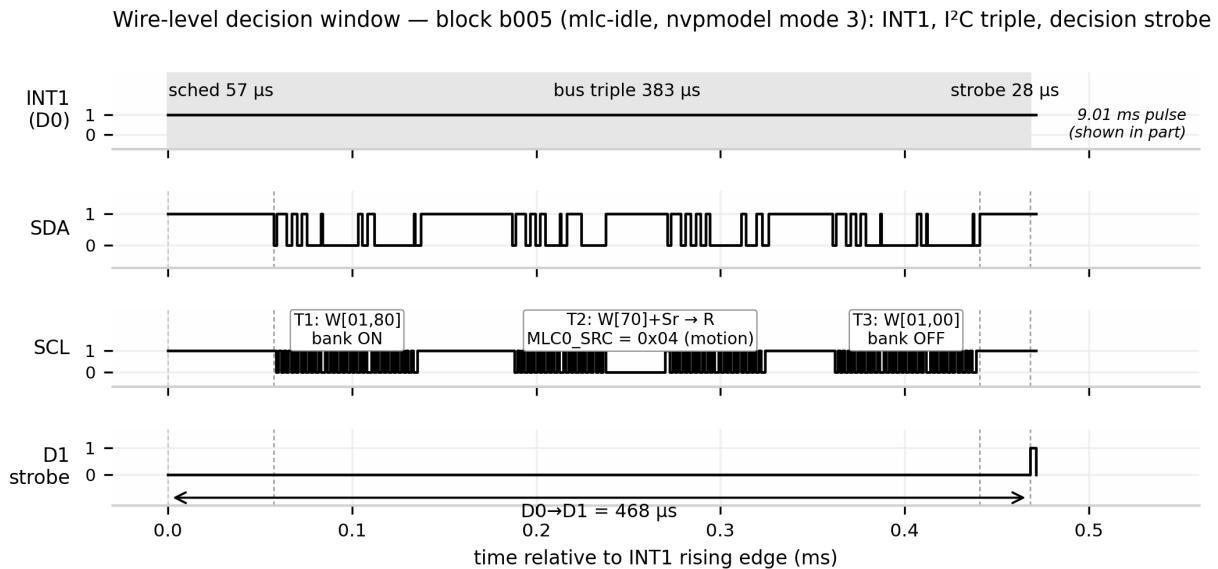


Figure 1: Measured wire-level decision window, confirmatory block b005

Fig. S4. Measured wire-level decision window for one representative trial (confirmatory block b005, mlc pipeline; Saleae capture at $t = 22.906$ s into the block). Lanes top to bottom: D0 (INT1), SDA, SCL, D1 (decision strobe). The three-transaction bank-switch read is decoded on SDA/SCL inside the D0→D1 interval: T1 writes `FUNC_CFG_ACCESS` (W 01 80), T2 reads `MLC0_SRC = 0x04`, T3 restores `FUNC_CFG_ACCESS` (W 01 00). For this trial: host scheduling 57.4 μs, I²C triple 383.4 μs, decision strobe 27.5 μs, total D0→D1 468.3 μs; no foreign I²C frames appear inside the window. Single-trial illustration, not a campaign statistic.